

Spatio-Temporal Correlation Between Precipitation and Vegetation Dynamics Within Brazil-Cerrado Region



Climatematch
Academy —

Lagoda Pod

Alex Adams

Isaac Yameogo

Joshua Solomon Avong

Julio Montenegro Gambini

Moses Kolleh Sesay

Contents

- Introduction
- Study Area
- Methodology
- Key Findings
- Discussion and Key Findings



Introduction

Scientific evidence suggests that **vegetation dynamics(health)** may be altering regional climate dynamics by **disrupting moisture recycling and atmospheric circulation**, potentially **intensifying local rainfall**. Empirical studies systematically linking of precipitations to spatial patterns of vegetation health remain limited.

“How has the frequency of precipitation in areas within Brazil (specifically Cerrado region), changed between 2001 - 2024, and how do these spatially correlate with vegetation dynamics (health)?”

Hypothesis: *A decrease in the frequency of precipitation in Brazil's Cerrado region between 2001 and 2024, will show a strong positive spatial correlation with a decline in vegetation health.*



Study Area

Over the past 20 years, Brazil has faced increasing environmental challenges at the intersection of climate extremes and ecosystem degradation. The rise in extreme daily precipitation events, leading to floods, landslides, and urban disruptions, has raised serious concerns for both urban and rural communities. Simultaneously, the country has experienced rapid forest loss, particularly in the Amazon and Cerrado biomes.



Methodology

1. Data processing
2. Geospatial Clipping
3. Defining Extremes
4. Trend and correlation analysis

Data Sources:

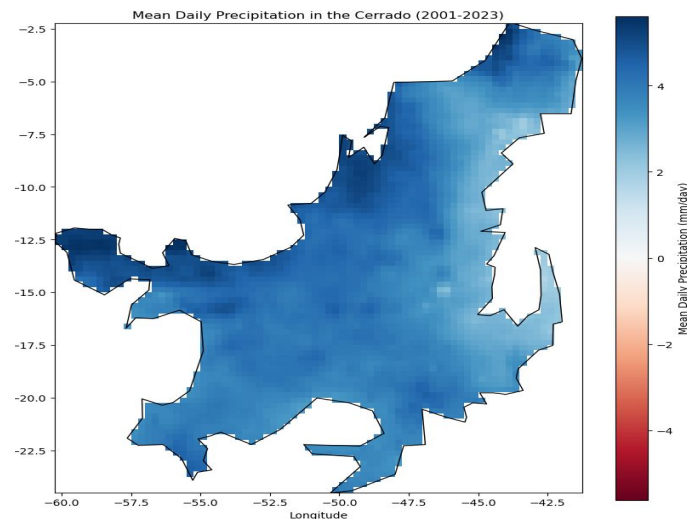
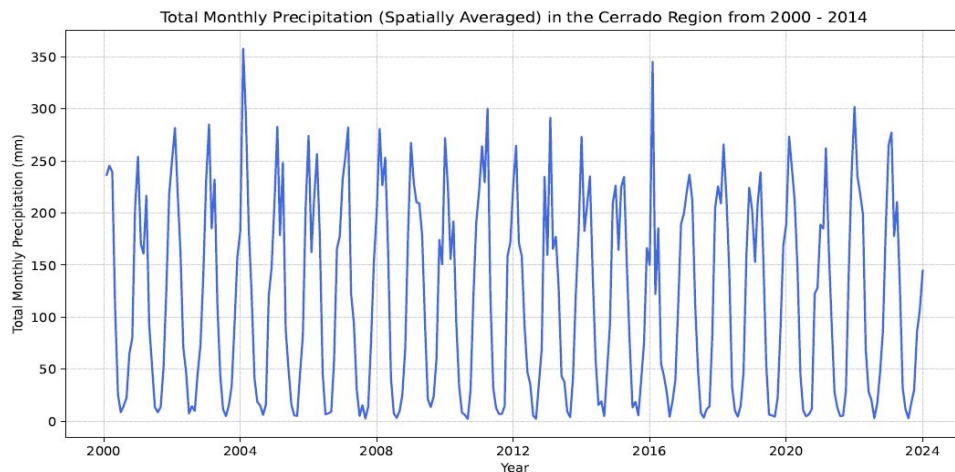
CHIRPS Precipitation Data: a high-resolution precipitation dataset combining satellite-derived precipitation estimates with ground-based station data to provide gridded precipitation data between 50°S-50°N.

NDVI Forest Loss Data: The **normalized difference vegetation index** is a widely used metric for quantifying the health/density of vegetation using sensor data. It is calculated from spectrometric data at two specific bands: red and near-infrared.



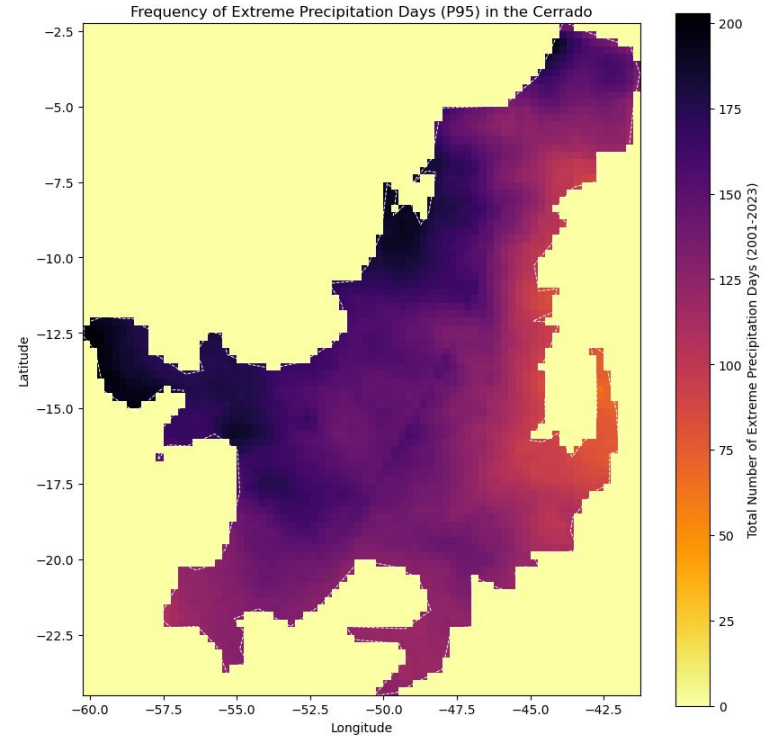
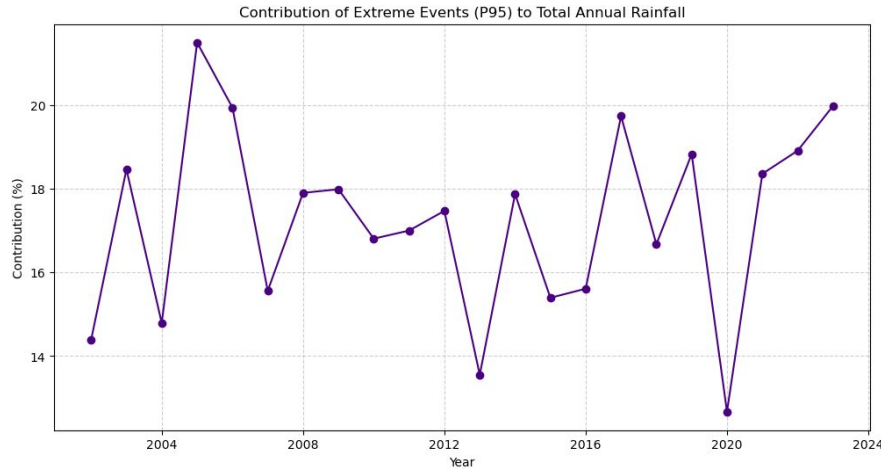
Key Findings

- Long term average precipitation shows that the northern is wetter than the southern part.
- Total monthly precipitation is varies by season for each of the years.



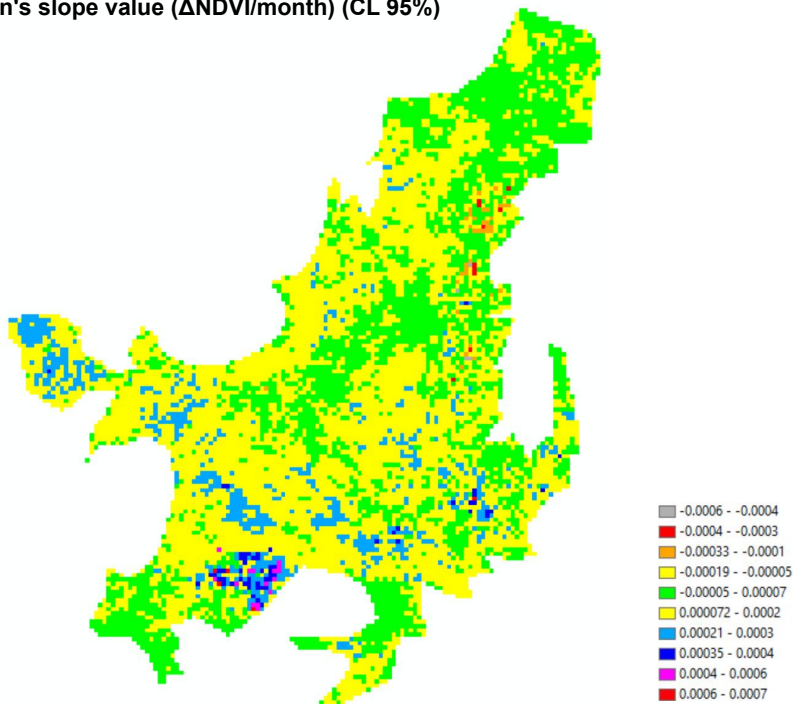
Results

- Hotspots: Some northern and central regions within the biome experience more frequent extreme precipitation events.

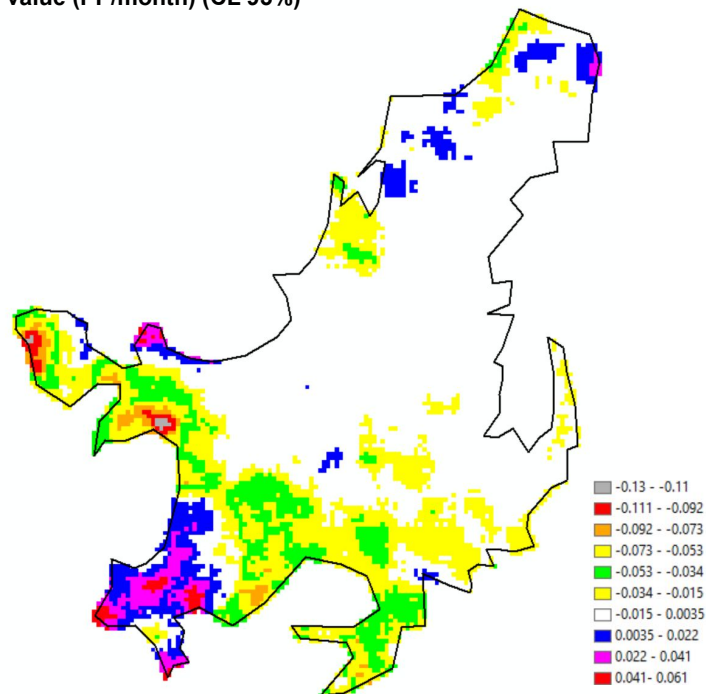


Results

Sen's slope value (Δ NDVI/month) (CL 95%)

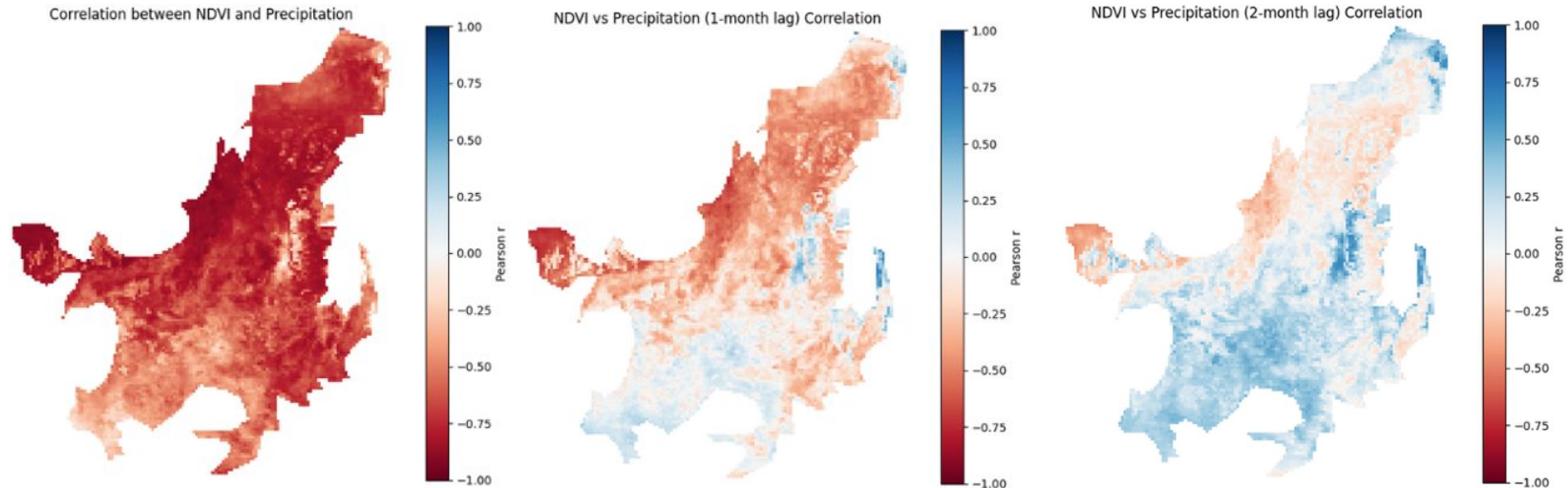


Sen's slope value (PP/month) (CL 95%)



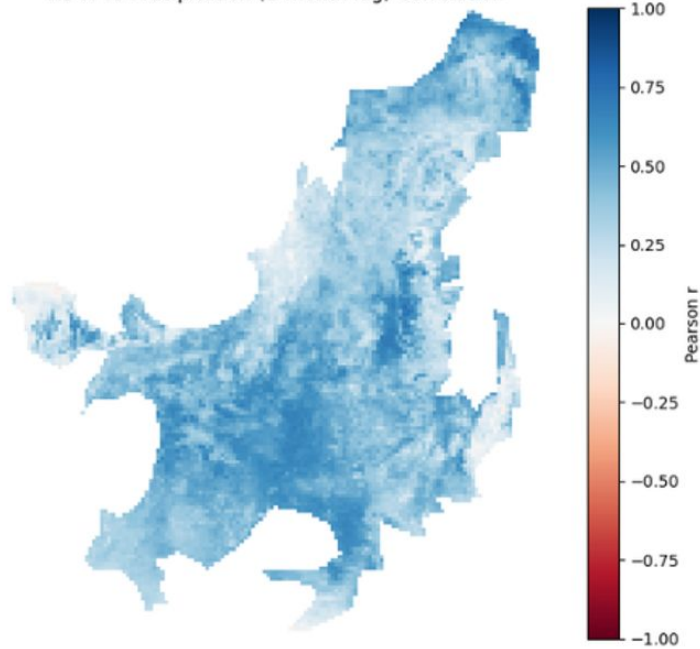
Key Findings

- Immediate negative link in the 0-1 month lag.
- Delayed positive response in the 2-4 months lag. This reveals the biome ecological memory; it relies on store water not immediate rainfall.

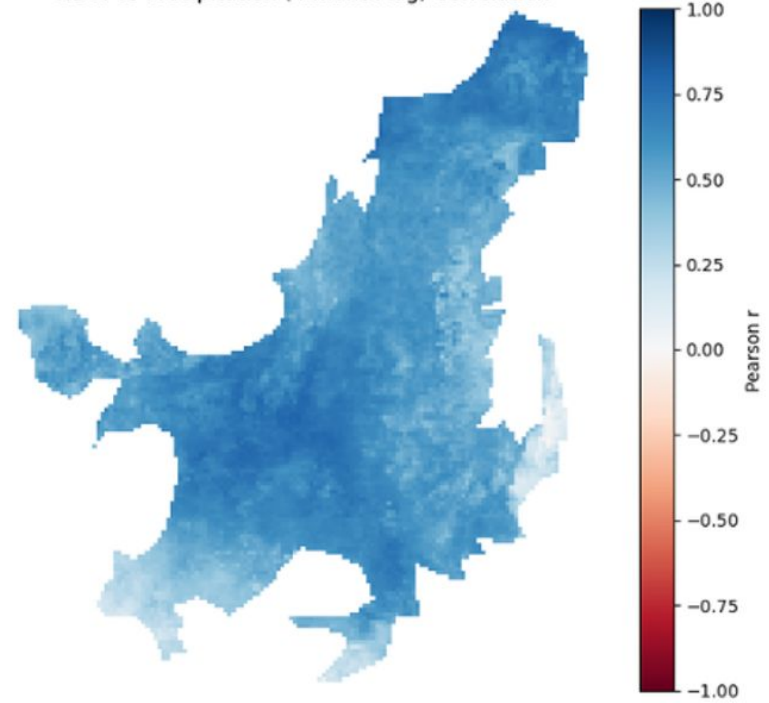


Key Findings

NDVI vs Precipitation (3-month lag) Correlation

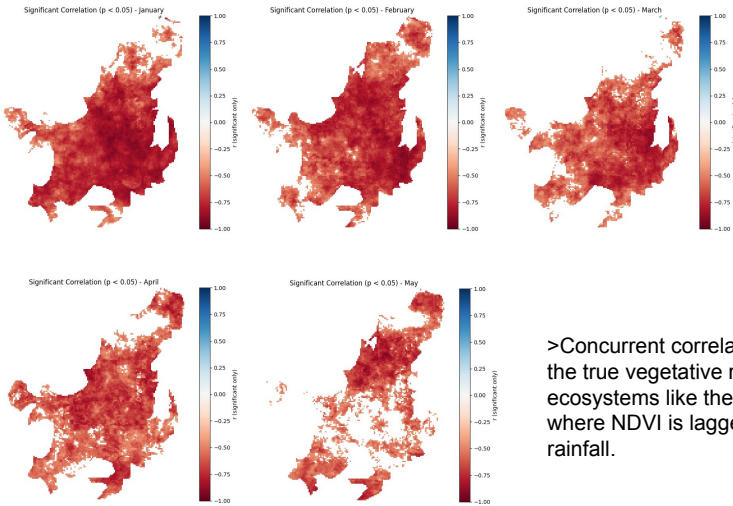


NDVI vs Precipitation (4-month lag) Correlation



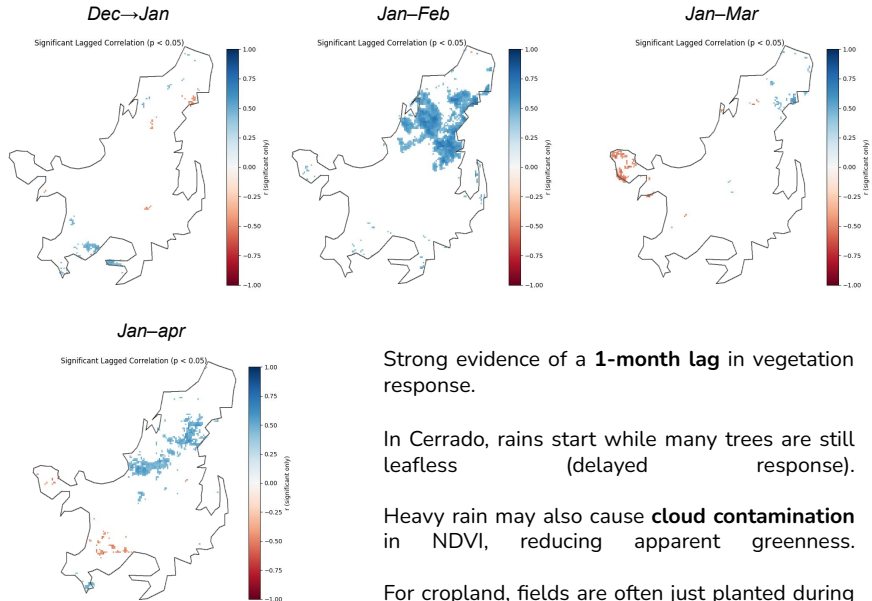
Key Findings

Concurrent correlation of monthly PP vs NDVI (Conf. level: 95%)



>Concurrent correlations miss the true vegetative response in ecosystems like the Cerrado where NDVI is lagged behind rainfall.

Lagged correlation of monthly PP vs NDVI (Conf. level: 95%)



Strong evidence of a **1-month lag** in vegetation response.

In Cerrado, rains start while many trees are still leafless (delayed response).

Heavy rain may also cause **cloud contamination** in NDVI, reducing apparent greenness.

For cropland, fields are often just planted during early rains → still low NDVI.

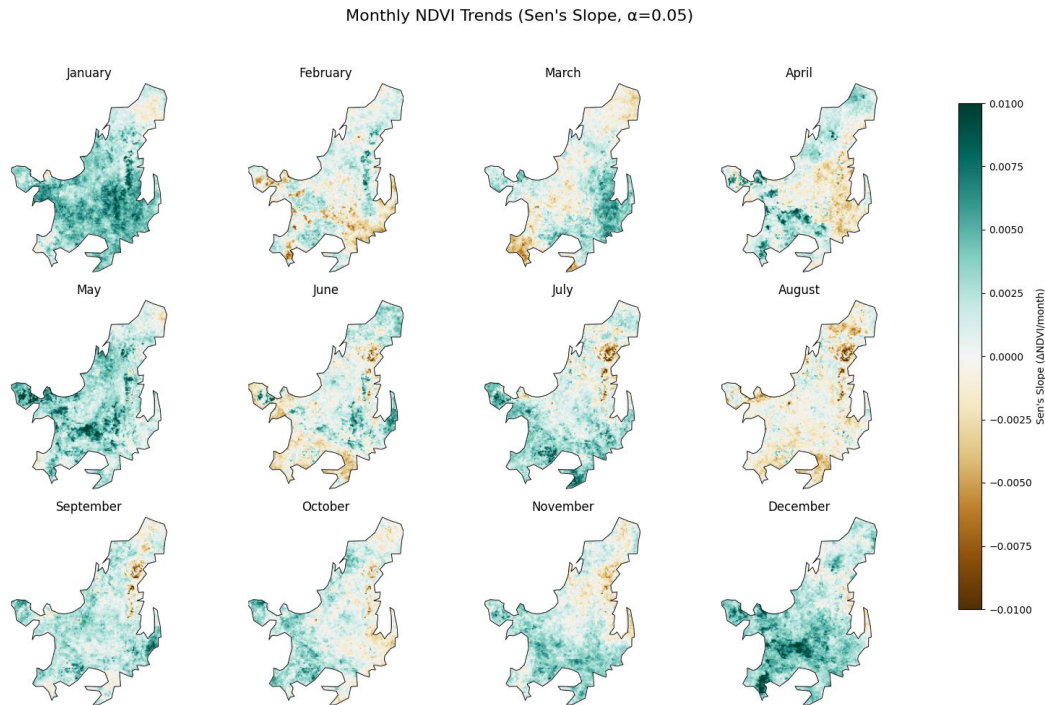
Key Findings

>**January:** Widespread **positive** NDVI trends.

>**July:** Peak dry season stress + **fire activity** increase in degraded/converted areas.

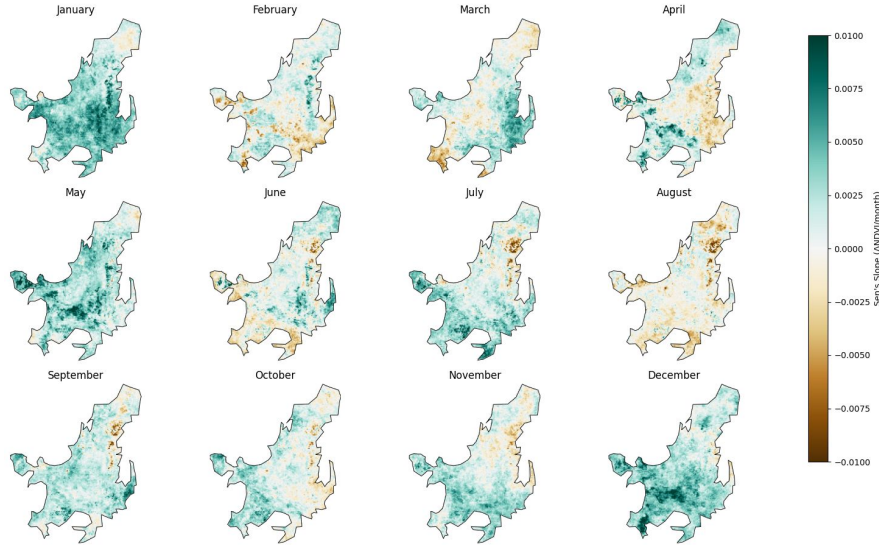
>**August:** likely reflects: land clearing, fire recurrence, overgrazing or pasture desiccation.

>**September:** Some regions begin greening, others still degrade. May reflect early rains or burned area recovery.



Key Findings

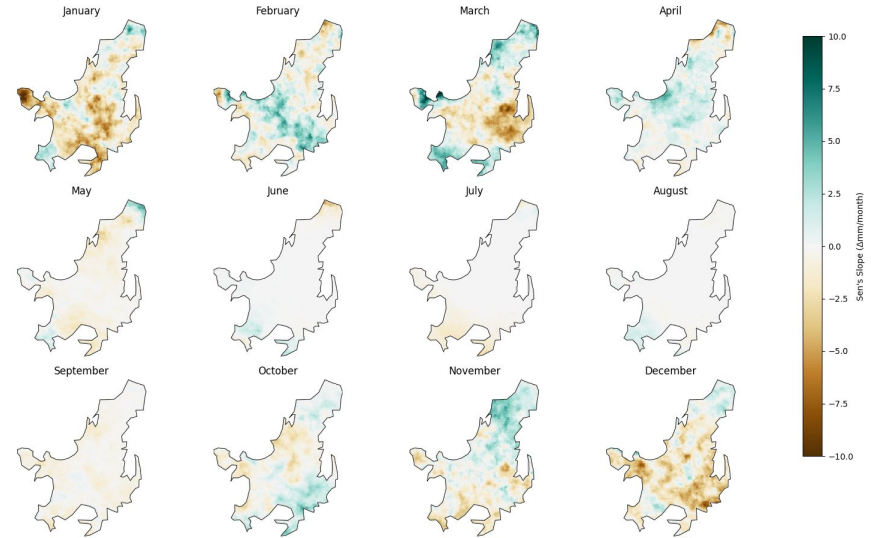
Monthly NDVI Trends (Sen's Slope, $\alpha=0.05$)



>**June-August:** NDVI decline is not climate-driven, likely caused by land use (e.g., fire, grazing, harvest). Lack of rain during dry season is expected; no trend change means NDVI decline (worsening stress or degradation).

> **July:** early signs of browning, **August:** strong degradation

Monthly Precipitation Trends (Sen's Slope, $\alpha=0.05$)



>**January:** Strongly spatial decline in P and surprisingly greening. Possibly adaptation or land-use intensification (e.g., earlier planting, irrigation). Greening despite less rain may signal shifts in phenology or land management rather than a climate-driven response.

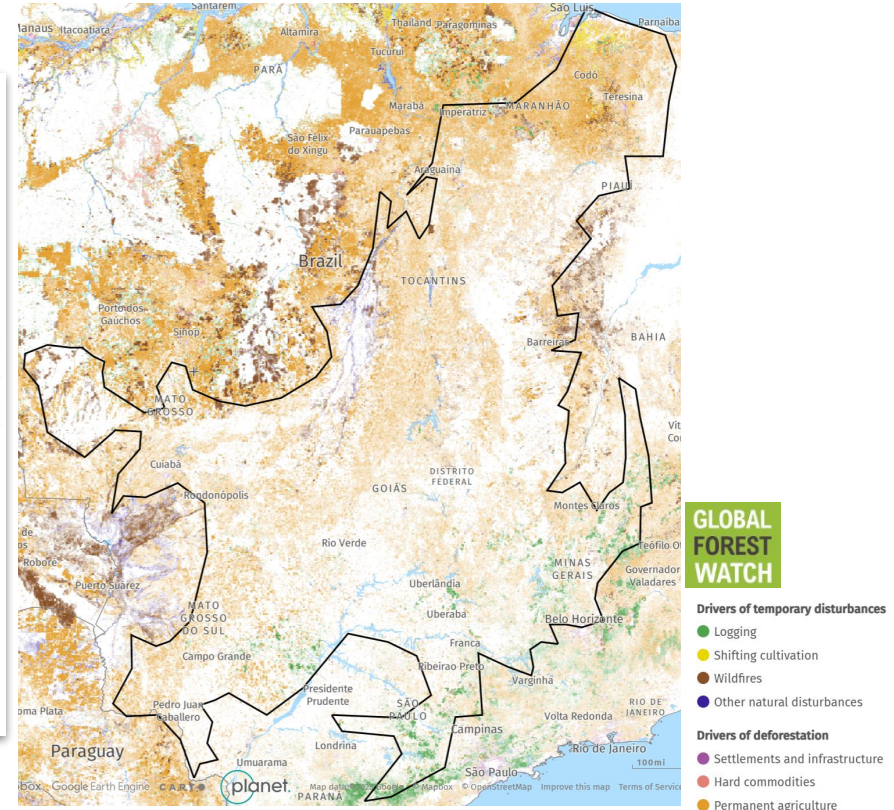
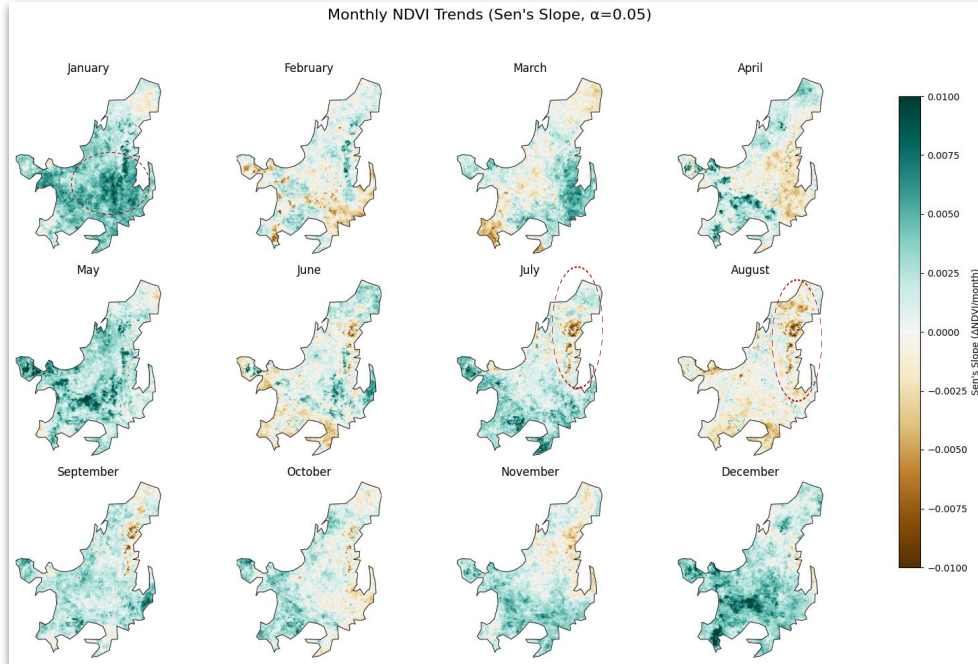


Key Findings

Month	Rain Trend	NDVI Trend	Interpretation
Jan	↓ Decreasing	↑ Increasing	Vegetation adapting or land-use response
Feb–Mar	Mixed	Mixed	Weak NDVI-rain coupling
Apr–May	Stable/mixed	↑ Strong greening	NDVI peaks slightly lagged from rain
Jun–Aug	~Stable	↓ Decline	NDVI decline due to degradation, not rainfall
Sep–Oct	Neutral	Mixed	Early greening in some areas, land use impact
Nov–Dec	↑ Increasing	↑ Greening	Onset of rains aligns with NDVI response



Key Findings



Future Context

- **Zonal analysis by land cover:** Identify where greening is **natural vs. anthropogenic** (e.g., pasture vs. savanna).
- **Classify adaptive zones:** Where NDVI trends are stable or increasing **despite decreasing precipitation**, which may be key to **climate resilience**
- Explore **Extreme** Rainfall Indices (e.g., Consecutive Dry Days, rainfall intensity, etc) for a fuller understanding of rainfall changes.
- Investigate **Physical Mechanisms:** Use reanalysis datasets (e.g., ERA5) to explore atmospheric drivers (e.g., moisture recycling, wind patterns, evapotranspiration) behind observed trends.
- Compare with Climate Model Projections: Evaluate observed trends against climate model outputs (e.g., CMIP6) to distinguish local vs. global climate influences.



THANK YOU FOR LISTENING!

Lagoda Pod - Danny McCulloch (TA)

Alex Adams

Isaac Yameogo

Joshua Solomon Avong

Julio Montenegro Gambini

Moses Kolleh Sesay



Future Context

- Trend in extreme precipitation frequency
- Extreme precipitation events are not random, they are concentrated in the wetter northern half of the biome.

